USER GUIDE

Labour Income Tax

A bracketed tax bends the constraint between consumption and leisure. Marginal rates kink the frontier; average rates break it, and crossing a threshold leaves the worker strictly worse off.

BEFORE YOU START

Opening it

There is nothing to install. It is a web page: it opens in Chrome, Safari, Edge or Firefox, on a computer, tablet or phone. You need a connection the first time; after that the browser keeps it and it works offline.

Top right there are two pairs of buttons. The first switches the language between **ES** and **EN**; the second goes from **Dark** to **Light**. For projecting in class, light mode reads better on a beamer; on screen, dark is easier on the eyes.

> **Every graph zooms.** Mouse wheel over the graph, or a two-finger pinch on the trackpad or touchscreen. The zoom applies to the graph under the pointer only, not to the whole page.

> Each slider has a number box beside it. For an exact value, type it into the box and press `Enter`; the slider is for exploring, the box is for precision.

THE MODEL, ON ONE PAGE

What the worker is choosing

The time endowment is normalised to **1**. The worker splits that unit between labour L and leisure $\ell = 1 - L$. Working the whole endowment earns **w**; working half of it, w/2. On top of that there is non-labour income $\mathbf{m_0}$, received whether he works or not.

Consumption is what is left after tax, divided by the price of the good: $c = (m_0 + w \cdot L - \text{tax}) / p_x$. The tax falls on labour income $w \cdot L$, not on m_0.

> That m_0 is untaxed is a modelling choice, and worth saying out loud in class: if it were taxed, the whole frontier would shift and the thresholds would land elsewhere. Here both modes use the same tax base, which is what makes them comparable.

The three brackets

Two thresholds, s_0 and s_1, defined on labour income, and three rates, t_0, t_1 and t_2. What matters is not the numbers but how they apply, and that is what the **How it applies** group decides.

MODE	HOW THE TAX IS WORKED OUT
Marginal rates	Each rate applies only to the slice of income falling inside its own bracket. This is how nearly every real system works. Net income is **continuous**: the frontier kinks but does not break, and earning one more always leaves you with more.
Average rates	The bracket you land in sets the rate on **all** of your income. At each threshold net income jumps down. Earning one more can leave you with considerably less.

WHAT IS ON SCREEN

The two panels

PANEL	WHAT IT SHOWS
Consumption and leisure	The constraint between consumption and leisure, with the utility map behind, the indifference curve through the optimum and the thresholds marked. The axis switch puts **labour** on the horizontal instead of leisure: the same picture mirrored, but easier to read when talking about hours worked.
Net income / **Utility**	The second panel shows either net against gross income — where the average-rate drop is unmissable — or utility traced along the frontier.

The readouts

READOUT	HOW TO READ IT
Consumption, Labour L*, Leisure	The optimal choice.
u(c*, ℓ*)	The utility it reaches.
Gross income	w·L*, before tax.
Tax	What is paid.
Average rate	Tax divided by gross income.
Marginal rate	What the taxman takes from the next unit earned. This is the number that governs the decision to work one more hour.
Kind of optimum	Tangency, kink, jump edge, no work or all work.
Largest drop	The biggest step in the frontier. Zero under marginal rates.

THE CENTRAL EXPERIMENT

The same brackets, two ways of applying them

This is the exercise that justifies the whole applet. The numbers are checked: if yours differ, make sure every slider is set.

1. Pick **Cobb–Douglas** with α = **0.5**. Wage and income: w = **100**, m_0 = **0**, p_X = **1**.
2. Brackets: t_0 = **0**, s_0 = **40**, t_1 = **0.2**, s_1 = **70**, t_2 = **0.8**. The first 40 is untaxed; between 40 and 70 the rate is 20 %; above that, 80 %.
3. Set it to **Marginal rates**.

   > L* = 0.45 — he works 45 % of his time. Consumption 44, leisure 0.55, u = 4.919. Gross income 45, tax 1. The applet calls it a **tangency**: an ordinary interior optimum.

4. Change only the mode to **Average rates**. Touch nothing else: same brackets, same wage, same preferences.

   > L* = 0.40 — he works less. Consumption 40, leisure 0.60, u = 4.899, and the optimum type becomes **jump edge**. He has parked himself just below the threshold.

5 Set the second panel to **Net income** and look at the step at 40.

```
At gross 40 he keeps 40; at 40.01 he keeps 32.01. One more
unit of gross income costs eight of net income.
```

6 Go back to **Marginal rates** and raise t_1 to **0.4**.

```
Now L* = 0.40 too, with u = 4.899, but the optimum type is
kink, not jump. There is bunching at the threshold here as
well, only for a gentler reason.
```

> That is the distinction to take away: **kinks** bunch people because the net wage falls abruptly, and **jumps** bunch them because crossing the threshold leaves them strictly worse off. The first is an efficiency cost; the second is simply a design error.

MORE TO TRY

Three variations

1 **Raise non-labour income.** With the same brackets in marginal mode, raise m_0 from 0 to 20. The worker gets richer without working more.

```
L* falls: with leisure a normal good, part of the extra income
is spent on not working.
```

2 **A confiscatory rate.** Raise t_2 to 0.95 and lower s_1 to 45.

```
Almost nobody crosses the second threshold: the frontier goes
nearly flat beyond it and working more barely pays.
```

3 **Quasilinear preferences.** Pick **Quasilinear** and compare the two modes again. In the custom box, x is consumption and y is leisure.

```
The income effect drops out of the linear good, so the
bunching shows more cleanly.
```

The layers

LAYER	WHAT IT DRAWS
Utility map	The coloured background.
Contours	Indifference curves at regular levels.
Feasible set	What is reachable under the frontier.
Curve through the optimum	The indifference curve of the solution.
Optimum	The chosen point.
Thresholds	The vertical lines at s_0/w and s_1/w. With these on you can see the optimum land right on top of one.
Grid	The grid.

WRITING YOUR OWN FUNCTION

The syntax it accepts

In the custom utility box, **x is consumption** and **y is leisure**. It is the applet's most confusing convention, so it is worth saying twice.

CONTROL	WHAT IT DOES
`+ − * / ^`	Add, subtract, multiply, divide, power. `x^0.5` is the square root of x.
`( )`	Brackets. `(x+y)^2` is not the same as `x+y^2`.
`sqrt ln log exp abs`	Root, natural log, base-10 log, exponential and absolute value.
`min max`	Two arguments: `min(x,y)` is perfect complements.
`sin cos`	Trigonometric, in radians. Rarely useful here, but available.
`e pi`	The two constants. `e` is 2.718…, `pi` is 3.1416…
`2x`, `3xy`	Implicit multiplication works: `2x` reads as `2*x`.

Examples that work

EXPRESSION	WHAT PREFERENCES IT DESCRIBES
`x^0.5*y^0.5`	Consumption and leisure weighted equally.
`x^0.7*y^0.3`	Someone who cares more about consuming: works more.
`x+2ln(y)`	Quasilinear in consumption. Optimal leisure does not depend on m_0.
`2ln(x)+y`	Quasilinear in leisure, the other way round.
`min(x,3y)`	Complements: a kink that interacts with the tax's own.

> If it cannot be read you get *"I can't read that expression"* and the graph stays as it was. The usual causes are an unclosed bracket, a decimal comma instead of a point (`0,5` instead of `0.5`) or a variable other than x and y.

IF SOMETHING GOES WRONG

Common problems

SYMPTOM	WHAT TO DO
The optimum comes out as "No work"	With a high m_0 or a low w that can be right. Raise the wage or lower non-labour income.
The thresholds do not appear	They are out of range: you need $0 < s/w < 1$, that is, thresholds below what full-time work earns. Lower s_0 and s_1, or raise w.
I see no difference between the two modes	With $t_0 = t_1 = t_2$ they coincide by construction. Spread the rates apart.
"Largest drop" comes out zero	You are in marginal rates, where net income is continuous by definition. That is the right answer.
The picture is mirrored from what I expect	Check the axis: **Leisure** and **Labour** give mirror images. Textbooks use leisure; when talking about hours, labour is usually clearer.

Open the applet: Labour Income Tax. It all runs in the browser, and once loaded, offline too.

← Back to the menu